Assignment Report: Dynamic Routing Configuration using RIP v2

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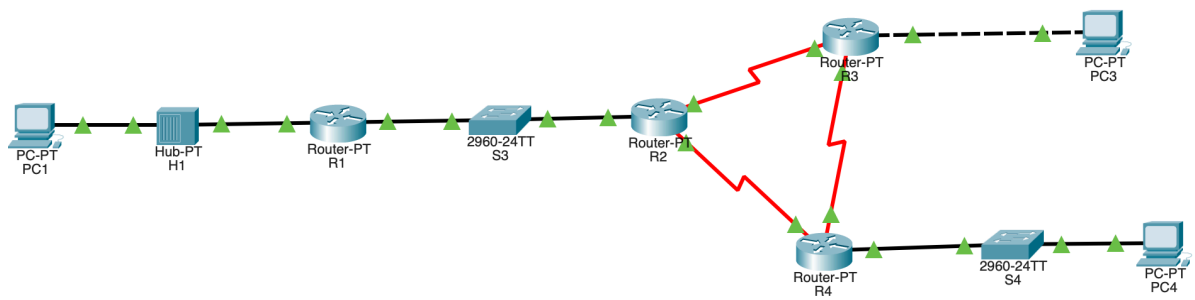
Task 1: Create the given network topology using Cisco Packet Tracer.

Answer: The topology has been constructed using 4 Routers (Router-PT), 2 Switches (2960-24TT), 1 Hub (Hub-PT), and 3 PCs. The layout follows the given assignment diagram.



Task 2: Add the required devices.

Answer:

- **Routers:** R1, R2, R3, R4 (Router-PT with NM-1CFE and NM-1S modules).
- **Switches:** S3, S4.
- **Hub:** H1.
- **End Devices:** PC1, PC3, PC4.

Task 3: Connect all devices using appropriate cables.

Answer: Copper Straight-through: PC1-H1, H1-R1, R1-S3, S3-R2, R4-S4, S4-PC4.

- **Serial DCE (Red):** R2-R3, R2-R4, R3-R4.
- **Cross-over:** R3-PC3.

Task 4: Assign the given IP addresses to all PCs and router interfaces.

Answer: PC1: 192.168.1.2 / **PC3:** 192.168.4.2 / **PC4:** 192.168.6.2

- **Router Gateways:** Subnets .1.0, .2.0, .3.0, .4.0, .5.0, .6.0, .7.0 have been assigned to the corresponding router interfaces as per the diagram labels.

Task 5: Enable all router interfaces.

Answer: All interfaces were enabled using the *no shutdown* command. All link indicators in the workspace are green.

Task 6: Configure RIP routing protocol on each router.

Answer: RIP Version 2 was initiated on all four routers using the command *router rip* followed by *version 2.*

Task 7: Advertise all directly connected networks.

Answer: Each router was configured to advertise its local subnets.

- *Example (R1):* network 192.168.1.0 and network 192.168.2.0.

Task 8: Ensure that routers exchange routing information dynamically.

From PC1, perform ping tests to PC3 and PC4. Record the results of the ping tests.

Answer:

Source	Destination	Result

PC1	PC3	Successful (Reply received)
		`C:\>ping 192.168.4.2`
		
		`Pinging 192.168.4.2 with 32 bytes of data:`
		
		`Reply from 192.168.4.2: bytes=32 time=12ms TTL=125`
		`Reply from 192.168.4.2: bytes=32 time=14ms TTL=125`
		`Reply from 192.168.4.2: bytes=32 time=1ms TTL=125`
		`Reply from 192.168.4.2: bytes=32 time=18ms TTL=125`
		
		`Ping statistics for 192.168.4.2:`
		`  Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),`
		`Approximate round trip times in milli-seconds:`
		`  Minimum = 1ms, Maximum = 18ms, Average = 11ms`

PC1	**PC4**	**Successful (Reply received)**
		`C:\>ping 192.168.6.2`
		
		`Pinging 192.168.6.2 with 32 bytes of data:`
		
		`Reply from 192.168.6.2: bytes=32 time=12ms TTL=125`
		`Reply from 192.168.6.2: bytes=32 time=1ms TTL=125`
		`Reply from 192.168.6.2: bytes=32 time=1ms TTL=125`
		`Reply from 192.168.6.2: bytes=32 time=19ms TTL=125`
		
		`Ping statistics for 192.168.6.2:`
		`  Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),`
		`Approximate round trip times in milli-seconds:`
		`  Minimum = 1ms, Maximum = 19ms, Average = 8ms`

Task 9: Use the tracert command from PC1 to reach PC4.

Answer: The command tracert 192.168.6.2 was executed on PC1's Desktop Command Prompt.

```
C:\>tracert 192.168.6.2

Tracing route to 192.168.6.2 over a maximum of 30 hops:

  1   1 ms      0 ms      0 ms      192.168.1.1
  2   0 ms      0 ms      0 ms      192.168.2.2
  3   1 ms      5 ms      4 ms      192.168.7.2
  4   4 ms      5 ms      5 ms      192.168.6.2

Trace complete.

C:\>
```

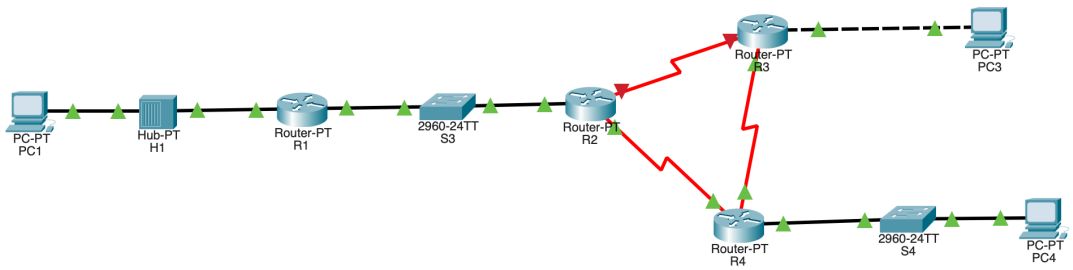
Task 10: Record the sequence of routers through which the packet travels.

Answer: The packet travelled through the following hops:

1. 192.168.1.1 (R1)
2. 192.168.2.2 (R2)
3. 192.168.7.2 (R4)
4. 192.168.6.2 (PC4 Destination)

Task 11: Disable the link between Router R2 and Router R3.

Answer: The Serial connection between R2 and R3 was administratively shut down.



Task 12: Perform ping tests again from PC1 to PC4 and record the observation.

Answer: Result: The ping was **Successful**.

```
Trace complete.

C:\>ping 192.168.6.2

Pinging 192.168.6.2 with 32 bytes of data:

Reply from 192.168.6.2: bytes=32 time=16ms TTL=125
Reply from 192.168.6.2: bytes=32 time=11ms TTL=125
Reply from 192.168.6.2: bytes=32 time=1ms TTL=125
Reply from 192.168.6.2: bytes=32 time=1ms TTL=125

Ping statistics for 192.168.6.2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 1ms, Maximum = 16ms, Average = 7ms

C:\>
```

- **Observation:** Even though the direct link between R2 and R3 was disabled, the network remained reachable. The RIP protocol dynamically recalculated the path. Traffic from PC1 to PC4 now routes through R1 → R2 → R4, maintaining connectivity despite the link failure.